MATH5916 Survival Analysis

Lecture 2: Parametric Survival Models

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Background

Confidence Intervals for Central Tendency

Weibull Survival Model

Comparisons for Parametric Models

Bathtub Hazard

Background

As discussed in the last lecture, the focus of a **survival analysis** is often centred around the survival or hazard functions

Estimating either $S(t)$ or $h(t)$ is very straightforward when there are **no censored observations**, but this is very rare

In this lecture, we will introduce parametric survival models for censored data with a focus on exponential and Weibull models

In later lectures, we will later discuss both non-parametric and semi-parametric models where the latter has aspects of both parametric and non-parametric methods

RECAP: Why aren't LMs / GLMs suitable for analysing survival data?

1. LMs / GLMs are not well-suited for survival data because they assume all events are observed.

2. LMs / GLMs are designed to model:

$$E(T|X) \rightarrow$$ Expected / mean survival time.

Through survival analysis, we have additional capability of estimating

$$S(t) \rightarrow$$ Probability of surviving beyond a certain point in time

and

Instantaneous risk of the event occurring $\leftarrow h(t)$ or $H(t) \rightarrow$ Cumulative hazard

All of the above functions provide deeper insight into the failure process compared to LMs / GLMs.

Exponential distribution for survival data

If the survival time is exponentially distributed, i.e., $T \sim Exp(\lambda)$ for $\lambda > 0$ and

$$f(t) = \lambda e^{-\lambda t}$$

$h(t+\delta)$

$= -\frac{d}{dt} \log S(t+\delta)$

then we already know the moments

$$\mathbb{E}(T) = \frac{1}{\lambda} \qquad \text{var}(T) = \frac{1}{\lambda^2}$$

and other important functions can be derived

$$S(t) = \mathbb{P}(T > t) = \int_t^\infty f(u)\,du = \int_t^\infty \lambda e^{-\lambda u}\,du = e^{-\lambda t}$$

$$h(t) = \frac{f(t)}{S(t)} = \lambda$$

$$\frac{P(T > t+\delta)}{P(T > t)} = P(T > \delta)$$

Exponential survival time has a constant hazard rate and the memoryless property, i.e.,

$$\mathbb{P}(T > t + \Delta t | T > t) = \mathbb{P}(T > \Delta t)$$

$S(t+\delta) = \boxed{S(t)}\, S(\delta)$

Likelihoods for survival data

Suppose we have a random sample of event times $T_1, T_2, \ldots, T_n$ each with probability density function $f(t_i|\theta)$. When all event times have been observed, i.e., no censoring, then we get the *usual* likelihood function for the parameter θ

$$\mathcal{L}(\theta) = \prod_{i=1}^{n} f(t_i|\theta)$$

That is, the contribution of each individual to the likelihood function is

$$\mathbb{P}(T_i = t_i|\theta) = \underline{\underline{f(t_i|\theta)}}$$

since we know their event times

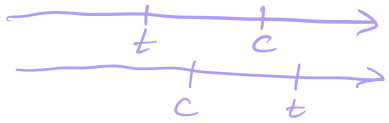
For censored individuals, however, we do not know their event times and $f(t_i|\theta)$ is no longer their contribution to the likelihood function

* What happens if we ignore censoring?

LIKELIHOODS FOR SURVIVAL DATA:

How do we construct the likelihood under right censoring?

Let $Y = \min(T, C)$



① Event is not observed $(\delta = 0)$:

$$P(Y=y, \delta=0 \mid C=c)$$

$$= P(Y=y \mid \delta=0, C=c)\, P(\delta=0 \mid C=c)$$

$$= P(Y=c \mid \delta=0, C=c)\, P(\delta=0 \mid C=c)$$

$$= 1 \times P(T>c)$$

$$= S(c)$$

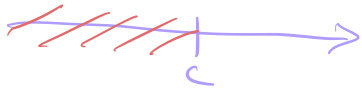
② Event is observed $(\delta = 1)$: $Y = \min(T, C)$

$$P(Y=y, \delta=1 \mid C=c)$$

$$= P(Y=y \mid \delta=1, C=c)\, P(\delta=1 \mid C=c)$$

$$= P(T=y \mid T \le c)\, P(\delta=1 \mid C=c)$$

$$= P(T=y \mid T \le c)\, P(T \le c)$$



$$= \frac{f(y)}{\cancel{P(T \le c)}}\, \cancel{P(T \le c)}$$

Likelihoods for survival data

For censored individuals, contributions to the likelihood function depend on the type of censoring, i.e., **what is known about the event time**

Right Censoring: the exact survival time is known to be greater than t

$$\mathbb{P}(T > t|\theta) = S(t|\theta)$$

Left Censoring: the exact survival time is known to be less than t

$$\mathbb{P}(T < t|\theta) = F(t|\theta) = 1 - S(t|\theta)$$

Interval Censoring: the exact survival time is known to have occurred in $t \in (t_\ell, t_u)$

$$\mathbb{P}(t_\ell < T < t_u|\theta) = S(t_\ell|\theta) - S(t_u|\theta)$$

Likelihoods for survival data

In the general case, where there are uncensored individuals and potentially all types of censoring, the likelihood function for the parameter θ is

$$\begin{aligned}\mathcal{L}(\theta) &= \prod_{\boxed{U}} \mathbb{P}(T_i = t_i|\theta) \prod_{RC} \mathbb{P}(T_i > t_i|\theta) \prod_{LC} \mathbb{P}(T_i < t_i|\theta) \prod_{IC} \mathbb{P}(t_\ell < T_i < t_u|\theta) \\ &= \prod_{U} f(t_i|\theta) \prod_{RC} S(t_i|\theta) \prod_{LC} (1 - S(t_i|\theta)) \prod_{IC} (S(t_\ell|\theta) - S(t_u|\theta))\end{aligned}$$

where U, RC, LC and IC denote individuals who were uncensored, right censored, left censored and interval censored, respectively

In this course, we will focus primarily on **right censoring** and will use C to denote this type of censoring

Exponential likelihood for right censoring

When there is only right censoring, the likelihood for a single sample is

$$\mathcal{L} = \prod_{U} f(t_i) \prod_{C} S(t_i) = \prod_{i=1}^{n} f(t_i)^{\delta_i} S(t_i)^{1-\delta_i}$$

where $t_1, \ldots, t_n$ are the observed follow-up times, and δ_i is an event indicator ($\delta_i = 1$ for failure and $\delta_i = 0$ for censoring)

For the exponential distribution with right censoring, the likelihood is

$$\begin{aligned}\mathcal{L}(\lambda) &= \prod_{U} \lambda e^{-\lambda t_i} \prod_{C} e^{-\lambda t_i} \\ &= \lambda^d \exp\left(-\lambda \sum_{i=1}^{n} t_i\right)\end{aligned}$$

where $d = \sum \delta_i$ is the number of uncensored observations, i.e., total deaths

Exponential Failure Time: MLE of λ

The log-likelihood is

$$\ell(\lambda) = \log \mathcal{L}(\lambda) = d \log \lambda - \lambda \sum_{i=1}^{n} t_i$$

with first derivative

$$\frac{d \log \mathcal{L}(\lambda)}{d\lambda} = \frac{d}{\lambda} - \sum_{i=1}^{n} t_i$$

and equating this to zero gives

$$\hat{\lambda} = \frac{d}{\sum_{i=1}^{n} t_i}$$

An estimated variance for $\hat{\lambda}$ can be derived using the asymptotic distribution of maximum likelihood estimators:

$$\widehat{\mathrm{var}}(\hat{\lambda}) = \left(-\mathbb{E}\left[\frac{d^2 \log \mathcal{L}(\lambda)}{d\lambda^2} \right] \Bigg|_{\lambda=\hat{\lambda}} \right)^{-1} = \frac{\hat{\lambda}^2}{d}$$

Example: 6-MP Leukaemia Trial

The results of fitting an exponential model to the remission times for leukaemia patients are as follows:

Group	d	$\sum t_i$	$\hat{\lambda}$	$\text{se}(\hat{\lambda})$
Placebo	21	182	0.1154	0.0252
6-MP	9	359	0.0251	0.0084

For example, the point and 95% interval estimates for λ in the 6-MP group are:

$$\hat{\lambda} = \frac{9}{359} = 0.0251$$

$$\text{se}(\hat{\lambda}) = \frac{9/359}{\sqrt{9}} = 0.0084$$

$$0.0251 \pm 1.96 \times 0.0084 = [0.0087, 0.0414]$$

Example: 6-MP Leukaemia Trial data in R

There are no R functions to estimate the exponential parameter directly, so we will first write a function to do so

```
exp.surv <- function(events, times){
  d <- sum(events)
  lhat <- d/sum(times)
  se.lhat <- lhat/sqrt(d)
  ci.lb <- lhat - qnorm(0.975)*se.lhat
  ci.ub <- lhat + qnorm(0.975)*se.lhat
  round(data.frame(lhat, se.lhat, ci.lb, ci.ub), digits = 4)
}
```

Next, we will read in the 6-MP data and separate them by treatment groups. In this data set, `time` is the follow-up time and `cens` is the event indicator

Example: 6-MP Leukaemia Trial data in R

```r
leukdat <- MASS::gehan
leukdat0 <- filter(leukdat, treat == "control")
leukdat1 <- filter(leukdat, treat == "6-MP")
exp.surv(leukdat0$cens, leukdat0$time)
```

```
##     lhat se.lhat ci.lb  ci.ub
## 1 0.1154  0.0252 0.066 0.1647
```

```r
exp.surv(leukdat1$cens, leukdat1$time)
```

```
##     lhat se.lhat  ci.lb  ci.ub
## 1 0.0251  0.0084 0.0087 0.0414
```

Survival Package in R

The `survival` package in R can be used to fit parametric models, but we should first start with a bit of background

Survival data can be thought of as having a bivariate outcome variable comprised of the observation time and an event indicator

To use the `survival` package, these two variables need to be combined and this is done using the `Surv` function

```
survival::Surv(time = leukdat1$time, event = leukdat1$cens)
```

```
##  [1] 10   7  32+ 23  22   6  16  34+ 32+ 25+ 11+ 20+ 19+  6  17+ 35+  6  13
## [20]  6+ 10+
```

Those with right censored event times now have a `+`

Survival Package in R

The `survreg` function in the `survival` package can be used to fit parametric models, we only need to specify `dist = "exponential"`

```
surv1 <- survreg(Surv(time, cens) ~ 1,
                 data = leukdat1,
                 dist = "exponential")
```

We can then use the generic `summary` function to gives us detailed results of this model

Example: 6-MP Leukaemia Trial data in R

```r
summary(surv1)
```

```
##
## Call:
## survreg(formula = Surv(time, cens) ~ 1, data = leukdat1, dist = "exponential")
##             Value Std. Error    z      p
## (Intercept) 3.686      0.333 11.1 <2e-16
##
## Scale fixed at 1
##
## Exponential distribution
## Loglik(model)= -42.2   Loglik(intercept only)= -42.2
## Number of Newton-Raphson Iterations: 4
## n= 21
```

Note that none of the outputted values correspond to our previous estimate $\boxed{\hat{\lambda} = 0.0251}$

Fitting parametric models in R

There is nothing wrong with the `survreg` function as the difference lies in the parameterisation of the model

The R function `survreg` fits a linear model to the log of the survival times

$$W = \log T = \mu + \sigma\epsilon$$

where μ is the intercept and σ the scale parameter

Different distributions for W (and hence T) can be obtained by choosing from a range of available distributions for the standardized variable

$$\epsilon = (W - \mu)/\sigma$$

For the 6-MP group using this model parameterisation, the estimate for λ corresponds to

$$\hat{\lambda} = \frac{1}{e^{\hat{\mu}}} = e^{-3.686} = 0.0250696$$

* For exponential / Weibull distribution on T, Survreg() uses standard extreme value errors.

Fitting parametric models in R

If we choose a least extreme value distribution for ϵ which has survival function

$$S_\epsilon(u) = \mathbb{P}(\epsilon > u) = \exp(-\exp(u))$$

then

$$\begin{aligned}
S_T(t) = \mathbb{P}(T > t) &= \mathbb{P}\left(\frac{\log T - \mu}{\sigma} > \frac{\log t - \mu}{\sigma}\right) \\
&= \mathbb{P}\left(\epsilon > \frac{\log t - \mu}{\sigma}\right) \\
&= S_\epsilon\left(\frac{\log t - \mu}{\sigma}\right) \\
&= \exp\left(-\exp\left(\frac{\log t - \mu}{\sigma}\right)\right) \\
&= \exp(-\exp(-\mu/\sigma)t^{1/\sigma})
\end{aligned}$$

Fitting parametric models in R

If we let

$$\lambda = \exp(-\mu/\sigma)$$
$$\gamma = 1/\sigma$$

then the survival function can be written as

$$\begin{aligned} S_T(t) &= \exp(-\exp(-\mu/\sigma)t^{1/\sigma}) \\ &= \underline{\underline{\exp(-\lambda t^{\gamma})}} \end{aligned}$$

Further, if the scale parameter σ is constrained to be 1, we get an exponential distribution for T with $\underline{\underline{\lambda = \exp(-\mu)}}$

For unconstrained σ, we get a Weibull distribution for T

Fitting parametric models in R

The basic syntax for fitting an exponential model is

```
survreg(Surv(times,events) ~ 1, dist = "exponential")
```

where `~ 1` is for an intercept only model

In the model output, R gives $\hat{\mu}$ and $\text{var}(\hat{\mu})$ which can be converted to $\hat{\lambda}$

$$\hat{\lambda} = \exp(-\hat{\mu})$$

But we also need its variance to conduct statistical inference

$$\text{var}(\hat{\lambda}) = \text{var}(\exp(-\hat{\mu}))$$

The delta method provides a means for deriving the asymptotic distribution of a transformation of the MLE

Delta method

Given a $p \times 1$ vector of parameter estimators $\hat{\boldsymbol{\theta}}$ with covariance matrix $\text{var}(\hat{\boldsymbol{\theta}})$, and a transformation of $\boldsymbol{\theta}$ to a $q \times 1$ parameter vector $\boldsymbol{\varphi} = g(\boldsymbol{\theta})$, the first-order Taylor series expansion of $g(\hat{\boldsymbol{\theta}})$ about $\boldsymbol{\theta}$ is

$$g(\hat{\boldsymbol{\theta}}) \approx g(\boldsymbol{\theta}) + \boxed{\left[\frac{\partial g_i(\boldsymbol{\theta})}{\partial \theta_j}\right](\hat{\boldsymbol{\theta}} - \boldsymbol{\theta})} \Rightarrow \text{Var}(g(\hat{\theta}) - g(\theta)) \approx \ldots$$

where $\left[\frac{\partial g_i(\boldsymbol{\theta})}{\partial \theta_j}\right]_{q \times p}$ is the Jacobian matrix of g with (i, j) element $\frac{\partial g_i(\boldsymbol{\theta})}{\partial \theta_j}$

Taking the variance of both sides of the equation gives us

$$\text{var}(\hat{\boldsymbol{\varphi}}) = \text{var}(g(\hat{\boldsymbol{\theta}})) \approx \left[\frac{\partial g_i(\boldsymbol{\theta})}{\partial \theta_j}\right] \text{var}(\hat{\boldsymbol{\theta}}) \left[\frac{\partial g_i(\boldsymbol{\theta})}{\partial \theta_j}\right]^T$$

Delta method

Evaluating the derivatives at the estimated value $\boldsymbol{\theta} = \hat{\boldsymbol{\theta}}$ gives the estimated covariance matrix

$$\widehat{\text{var}}(\hat{\boldsymbol{\varphi}}) = \widehat{\text{var}}(g(\hat{\boldsymbol{\theta}})) \approx \left[\frac{\partial g_i(\hat{\boldsymbol{\theta}})}{\partial \theta_j}\right] \widehat{\text{var}}(\hat{\boldsymbol{\theta}}) \left[\frac{\partial g_i(\hat{\boldsymbol{\theta}})}{\partial \theta_j}\right]^T$$

For the case $p = q = 1$,

$$\widehat{\text{var}}(\hat{\varphi}) \approx \left(\frac{dg(\hat{\theta})}{d\theta}\right)^2 \widehat{\text{var}}(\hat{\theta})$$

Delta method: Exponential Failure Time

In the exponential distribution example above, the transformation is

$$\lambda = g(\mu) = \exp(-\mu)$$

which has derivative

$$\frac{dg(\mu)}{d\mu} = -\exp(-\mu)$$

So, the estimated variance and standard errors are

$$\begin{aligned}
\widehat{\text{var}}(\hat{\lambda}) &\approx \left(\frac{dg(\hat{\mu})}{d\mu}\right)^2 \widehat{\text{var}}(\hat{\mu}) \\
&= (\exp(-\hat{\mu}))^2 \widehat{\text{var}}(\hat{\mu}) \\
\text{se}(\hat{\lambda}) &= \exp(-\hat{\mu})\text{se}(\hat{\mu})
\end{aligned}$$

Fitting parametric models in R

The estimated exponential model for the 6-MP data was

```r
summary(surv1)$table
```

```
##                 Value Std. Error        z            p
## (Intercept) 3.686098  0.3333333 11.05829 1.99864e-28
```

Using the delta method, the transformed values are

```r
exp(-surv1$coef) * c(1, sqrt(surv1$var))
```

```
## [1] 0.025069638 0.008356546
```

which directly correspond to our previous estimates, $\hat{\lambda} = 0.0251$ and $\text{se}(\hat{\lambda}) = 0.0084$

Fitting parametric models in R

We could also write an R function to do the conversion:

```
exp.conv <- function(mu, var.mu){
  lambda <- exp(-mu)
  var.lambda <- (lambda^2)*var.mu
  se.lambda <- sqrt(var.lambda)
  ests <- c(lambda,var.lambda,se.lambda)
  names(ests) <- c("coef","var","s.e.")
  round(ests, digits = 4)
}
exp.conv(surv1$coef, surv1$var)
```

```
##   coef    var   s.e.
## 0.0251 0.0001 0.0084
```

Background

Confidence Intervals for Central Tendency

Weibull Survival Model

Comparisons for Parametric Models

Bathtub Hazard

Confidence intervals for parameters

There are several methods for constructing confidence intervals for λ. These include the exact method, asymptotic method and the likelihood ratio test.

An exact confidence interval exists for the exponential distribution when there is no censoring. In this case, the maximum likelihood estimator is

$$\hat{\lambda} = \frac{n}{\sum_{i=1}^{n} T_i}$$

where n is the number of observations and is fixed (i.e., non-random). Now, the distribution of $2n\lambda/\hat{\lambda}$ is

Gamma $\left(\frac{r}{2}, 2\right)$

$$\frac{2n\lambda}{\hat{\lambda}} = 2\lambda \sum_{i=1}^{n} T_i \sim \text{gamma}(n, 1/2) \equiv \chi^2_{2n}$$

where $2\lambda T_i \sim Exp(1/2)$ since $T_i \sim Exp(\lambda)$, and the T_i's are independent

Exact method

Using the fact that $2n\lambda/\hat{\lambda} \sim \chi^2_{2n}$, we get the probabilities

$$\mathbb{P}\left(\chi^2_{2n,\alpha/2} < \frac{2n\lambda}{\hat{\lambda}} < \chi^2_{2n,1-\alpha/2}\right) = 1-\alpha$$

$$\mathbb{P}\left(\frac{\hat{\lambda}\chi^2_{2n,\alpha/2}}{2n} < \lambda < \frac{\hat{\lambda}\chi^2_{2n,1-\alpha/2}}{2n}\right) = 1-\alpha$$

Therefore, an *exact* $(1-\alpha)100\%$ confidence interval for λ is

$$\left(\frac{\hat{\lambda}\chi^2_{2n,\alpha/2}}{2n}, \frac{\hat{\lambda}\chi^2_{2n,1-\alpha/2}}{2n}\right)$$

6-MP Leukaemia Trial data

For the placebo group, an exact 95% confidence interval for λ is

$$\left(\frac{0.1154 \times 26.0}{42}, \frac{0.1154 \times 61.8}{42}\right) = (0.071, 0.170)$$

The 6-MP group is subject to censoring, so the exact method is not available, but an approximate confidence interval can be obtained by treating $2d\lambda/\hat{\lambda}$ as a χ^2_{2d} random variable

This method gives as a 95% confidence interval for λ

$$\left(\frac{0.0251 \times 8.231}{18}, \frac{0.0251 \times 31.526}{18}\right) = (0.0115, 0.0439)$$

Asymptotic method

Asymptotically, $\hat{\lambda} \sim \mathcal{N}(\lambda, \lambda^2/d)$, so an approximate 95% confidence interval for λ is

$$\hat{\lambda} \pm 1.96 \times \text{se}(\hat{\lambda})$$

For the 6-MP group, this method gives

$$0.0251 \pm 1.96 \times 0.0084 = (0.0087, 0.0414)$$

Likelihood ratio test

The likelihood ratio statistic for testing $H_0 : \lambda = \lambda_0$ versus $H_1 : \lambda \neq \lambda_0$ is, under H_0,

$$W(\lambda_0) = 2(\log \mathcal{L}(\hat{\lambda}) - \log \mathcal{L}(\lambda_0)) \approx \chi^2_1$$

So, a 95% confidence interval for λ consists of all λ_0 such that $H_0 : \lambda = \lambda_0$ is not rejected at the 5% level, i.e.,

$$\log \mathcal{L}(\hat{\lambda}) - \log \mathcal{L}(\lambda_0) \leq \chi^2_{1,0.95}/2 = 1.92$$

This approach is sometimes referred to as profile likelihoods

$$\ell(\lambda) = \ell(\hat{\lambda}) + (\hat{\lambda}-\lambda)\,\underline{\underline{\ell'(\hat{\lambda})}} + \frac{1}{2}(\hat{\lambda}-\lambda)^2\,\ell''(\hat{\lambda}) + \dots$$

$$\ell(\lambda) - \ell(\hat{\lambda}) = (\hat{\lambda}-\lambda)\,\underbrace{\ell'(\hat{\lambda})}_{=0} + \frac{1}{2}(\hat{\lambda}-\lambda)^2\,\underline{\underline{\ell''(\hat{\lambda})}} + \dots$$

1 $$P\Big(\underbrace{2\big(\ell(\hat{\lambda}) - \ell(\lambda)\big) \le \underline{\underline{\chi^2_{1,0.95}}}}\Big) \approx \underline{\underline{0.95}}$$

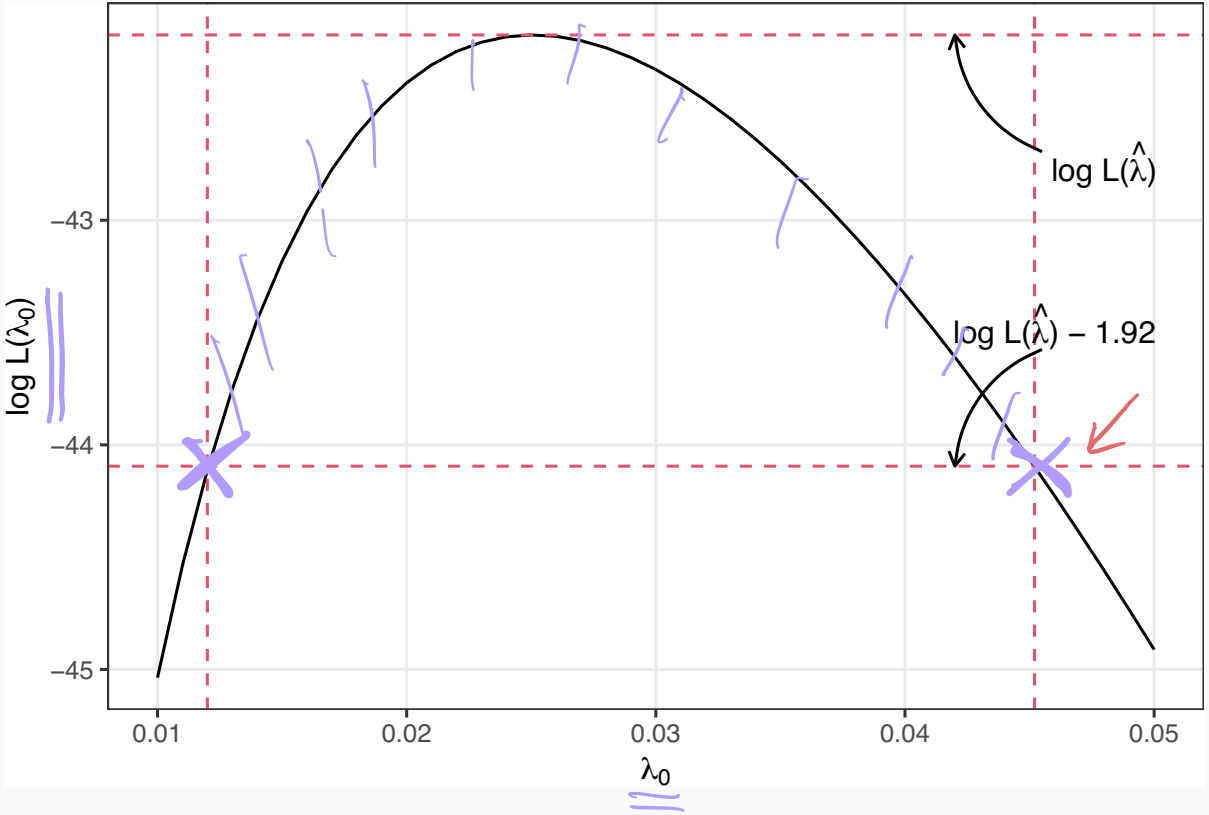
$$C(\underline{\text{data}}) = \Big\{\lambda : 2\big(\underbrace{\ell(\hat{\lambda})} - \underline{\underline{\ell(\lambda)}}\big) \le \chi^2_{1,0.95}\Big\}$$

6-MP Leukaemia Trial data

The procedure is illustrated graphically for the 6-MP group below, where the confidence interval derived from this method is (0.0121,0.0452) $2(\ell(\hat{\lambda}) - \ell(\lambda_0)) \leq 1.92 \times 2$

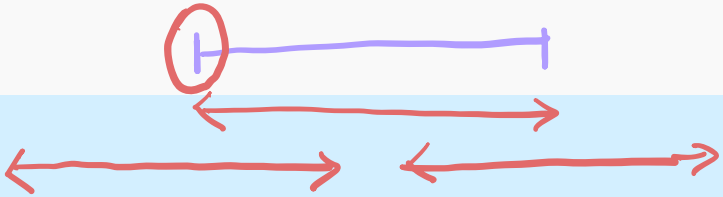
$\ell(\hat{\lambda}) - \ell(\lambda_0) \leq 1.92$

$\ell(\lambda_0) \geq \ell(\hat{\lambda}) - 1.92$



Fitting parametric models in R

The exact and asymptotic methods are easily computed in R. The LRT method, however, is much more involved



```
lrt.ci <- function(events,times,alpha=0.05){
  d <- sum(events)
  lhat <- d/sum(times)
  asym.lb <- lhat-qnorm(1-alpha/2)*lhat/sqrt(d)
  asym.ub <- lhat+qnorm(1-alpha/2)*lhat/sqrt(d)
  f <- function(lam) 2*(d*log(lhat/lam) - (lhat-lam)*sum(times)) - qchisq(1-alpha,1)
  ci.lb <- uniroot(f,lower=max(asym.lb-(asym.ub-asym.lb)/2,0),
                   upper=asym.lb+(asym.ub-asym.lb)/2)$root
  ci.ub <- uniroot(f,lower=asym.ub-(asym.ub-asym.lb)/2,
                   upper=asym.ub+(asym.ub-asym.lb)/2)$root
  round(data.frame(ci.lb,ci.ub), digits = 4)
}
lrt.ci(leukdat1$cens,leukdat1$time)
```

```
##    ci.lb  ci.ub
## 1 0.0121 0.0452
```

Median survival time

Since survival time often follows a skewed distribution, statistical inference about its mean is not usually conducted

The median survival time is the time prior to which 50% of the individuals in the population under study are expected to fail, and is given by the time $t_{0.5}$ such that

$$\mathbb{P}(T \leq t_{0.5}) = \boxed{1 - S(t_{0.5}) = 0.5}$$

More generally, the α-quantile of the distribution of survival times is given by t_α such that $\mathbb{P}(T \leq t_\alpha) = 1 - S(t_\alpha) = \alpha$

Median survival time for exponential

For exponentially distributed survival times, we have

$e^{-\lambda t}$

$$
\begin{aligned}
0.5 &= 1 - S(t_{0.5}) \\
&= 1 - e^{-\lambda t_{0.5}} \\
e^{-\lambda t_{0.5}} &= 0.5
\end{aligned}
$$

so that

$$t_{0.5} = \frac{\log 2}{\lambda}$$

For the 6-MP group, an estimate of the median survival time is then

$$\hat{t}_{0.5} = \frac{\log 2}{\hat{\lambda}} = \frac{\log 2}{0.0251} = 27.649 \text{ weeks}$$

Confidence intervals for median survival time

Since median survival is a simple transformation of the exponential parameter, confidence intervals can be obtained by transforming the confidence limits for λ

That is, if we have a 95% confidence interval (λ_L, λ_U) for λ such that $\mathbb{P}(\lambda_L < \lambda < \lambda_U) = 0.95$, then we also have

$$\mathbb{P}\left(\frac{\log 2}{\lambda_U} < \frac{\log 2}{\lambda} < \frac{\log 2}{\lambda_L}\right) = 0.95$$

So, a 95% confidence interval for median survival $t_{0.5}$, assuming an exponential model, is

$$\left(\frac{\log 2}{\hat{\lambda}_U}, \frac{\log 2}{\hat{\lambda}_L}\right)$$

where $(\hat{\lambda}_L, \hat{\lambda}_U)$ is the 95% confidence interval for λ

Confidence intervals for median survival time

For the 6-MP group with 95% confidence intervals for λ based on Methods 1-3, the corresponding 95% confidence intervals for the median survival time are:

Method	95% CI
Exact	(15.786, 60.466)
Asymptotic	(16.723, 79.756)
LRT	(15.342, 57.679)

Confidence intervals for survival time quantiles

Confidence intervals for survival time quantiles can also be derived by making a distributional assumption about the estimated quantiles. Empirical evidence suggests that the log of the estimated quantile is approximately normally distributed.

For exponentially distributed survival times, the log qth quantile for survival time is

$P(T \le t_q) = q$

$1 - S(t_q) = q$

$1 - e^{-\lambda t_q} = q$

$$\log \hat{t}_q = \log\left(-\frac{\log(1-q)}{\hat{\lambda}}\right)$$
$$= \log(-\log(1-q)) - \log \hat{\lambda}$$

and using the delta method an approximate standard error is

$$\mathrm{se}(\log \hat{t}_q) = \frac{\mathrm{se}(\hat{\lambda})}{\hat{\lambda}}$$

Confidence intervals for survival time quantiles

Hence, a 95% confidence interval for $\log t_{0.5}$ is

$$\log \hat{t}_{0.5} \pm 1.96 \times \mathrm{se}(\hat{\lambda})/\hat{\lambda}$$

and converting to a 95% confidence interval for $t_{0.5}$ yields

$$\left(\hat{t}_{0.5} \exp\left(-1.96 \cdot \frac{\mathrm{se}(\hat{\lambda})}{\hat{\lambda}}\right), \hat{t}_{0.5} \exp\left(1.96 \cdot \frac{\mathrm{se}(\hat{\lambda})}{\hat{\lambda}}\right)\right)$$

For the 6-MP group, this method gives a 95% confidence interval for $t_{0.5}$ as

$$27.649 e^{\pm 1.96 \times 0.0084/0.0251} = (14.386, 53.139)$$

Confidence intervals for survival time quantiles in R

The R function `predict` can compute quantiles and their confidence intervals

For example, the 0.25, 0.5, and 0.75 quantiles (and CI's) of the survival time for the 6-MP group (assuming log quantile is approximately normal)

```
qpred <- predict(surv1, data.frame(1), type = "uquantile",
                 p = c(0.25,0.5,0.75), se.fit = TRUE) %>%
  as.data.frame(row.names = c(0.25,0.5,0.75)) %>%
  mutate(ests = exp(fit),
         ci.lb = exp(fit - 1.96*se.fit),
         ci.ub = exp(fit + 1.96*se.fit))
qpred
```

```
##           fit    se.fit     ests     ci.lb     ci.ub
## 0.25 2.440198 0.3333333 11.47532  5.970706  22.05483
## 0.5  3.319585 0.3333333 27.64887 14.385943  53.13938
## 0.75 4.012732 0.3333333 55.29774 28.771886 106.27875
```

Hypothesis testing: comparing treatment groups

We can compare two different treatment groups on the basis of an assumed parametric model by including a treatment indicator in the log-linear model:

$$\log T_i = \mu + \alpha z_i + \sigma \epsilon_i$$

where z_i is the treatment group indicator: $z_i = 0$ if patient i belongs to group 0, $z_i = 1$ if patient i belongs to group 1. Then, a test of hypotheses

$$H_0 : \alpha = 0$$
$$H_1 : \alpha \neq 0$$

is equivalent to a test for treatment difference

6-MP Leukaemia Trial in R

```r
leukdat$treat <- factor(leukdat$treat, levels = c("control", "6-MP"))
survt <- survreg(Surv(time,cens) ~ treat, data = leukdat, dist = "exponential")
summary(survt)
```

```
##
## Call:
## survreg(formula = Surv(time, cens) ~ treat, data = leukdat, dist = "exponential")
##             Value Std. Error    z       p
## (Intercept) 2.159      0.218 9.90 < 2e-16
## treat6-MP   1.527      0.398 3.83 0.00013
##
## Scale fixed at 1
##
## Exponential distribution
## Loglik(model)= -108.5   Loglik(intercept only)= -116.8
##  Chisq= 16.49 on 1 degrees of freedom, p= 4.9e-05
## Number of Newton-Raphson Iterations: 4
## n= 42
```

6-MP Leukaemia Trial in R

The estimated hazard rates for the two groups are

$$\begin{aligned}\hat{\lambda}_{z_i} &= \exp(-(\hat{\mu} + \hat{\alpha} z_i)) \\ &= \exp(-2.159 - 1.527 \times z_i)\end{aligned}$$

For placebo patients, that is

$$\hat{\lambda}_0 = \exp(-\hat{\mu}) = e^{-2.159} = 0.1154$$

and for 6-MP patients

$$\hat{\lambda}_1 = \exp(-(\hat{\mu} + \hat{\alpha})) = e^{-(2.16-1.53)} = 0.0251$$

The Wald and likelihood ratio statistics are highly significant (e.g., the likelihood ratio chi-square statistic is 16.49 on 1 degree of freedom, p-value=0.000049). So, we reject the null hypothesis of no difference in survival experience between the treatment groups.

Background

Confidence Intervals for Central Tendency

Weibull Survival Model

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Bathtub Hazard

Weibull Distribution

The constant hazard rate assumption underlying exponentially distributed survival times can be unrealistic. The Weibull distribution extends the exponential distribution by adding a shape parameter.

The Weibull hazard function for parameters $\lambda, \gamma > 0$ is

$$h(t) = \lambda\gamma t^{\gamma-1}$$

and the usual, important functions can be derived from it

$$S(t) = \exp(-H(t)) = \exp\left(-\int_0^t h(u)\,du\right) = \exp(-\lambda t^{\gamma})$$

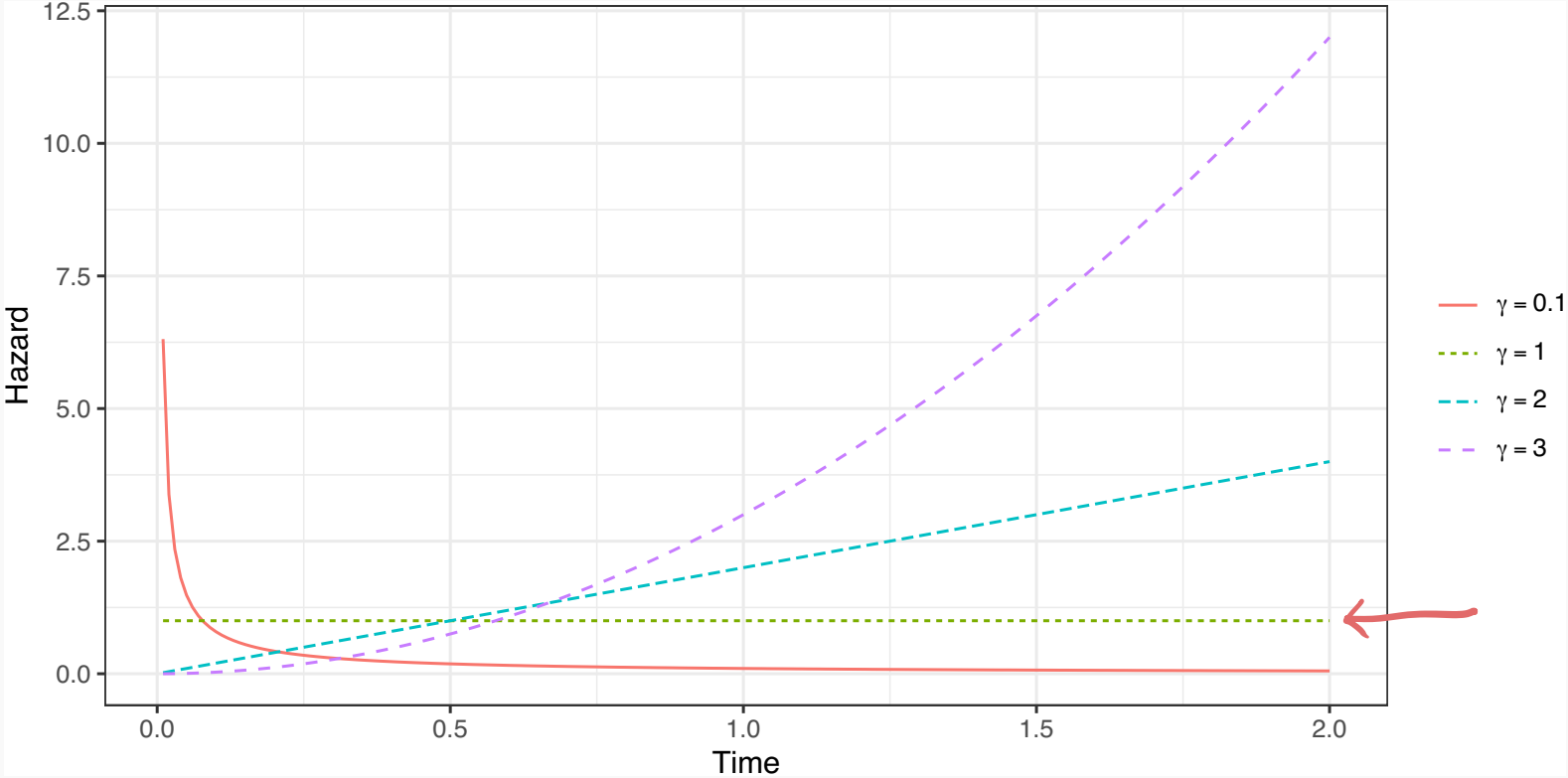
$$f(t) = h(t)S(t) = \lambda\gamma t^{\gamma-1}\exp(-\lambda t^{\gamma})$$

set this to 1 $\Rightarrow$ $S(t)$ for an exponential

where λ is the scale parameter and γ the shape parameter

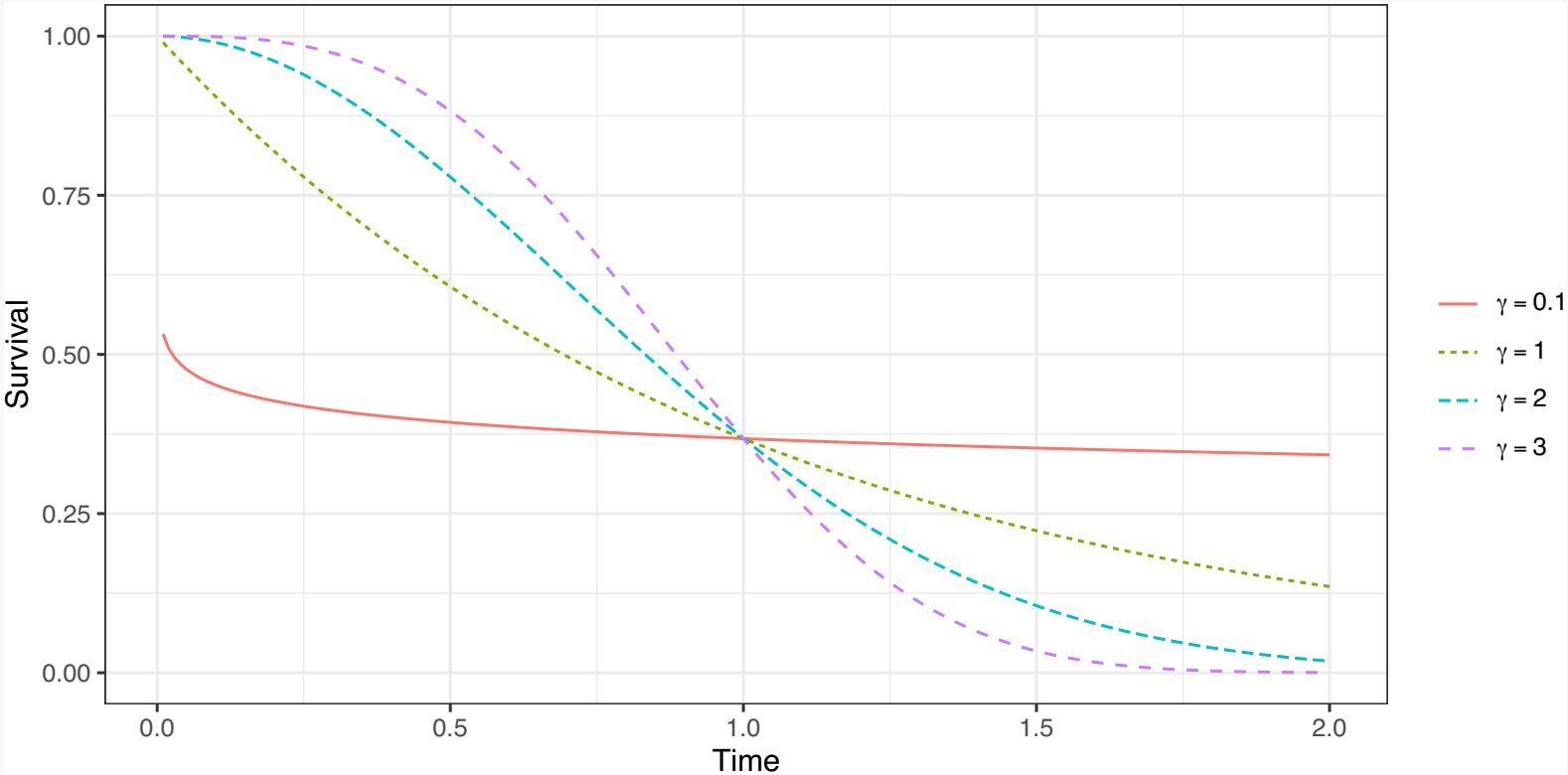
The Weibull simplifies to the exponential when $\gamma = 1$

Weibull Hazard Function

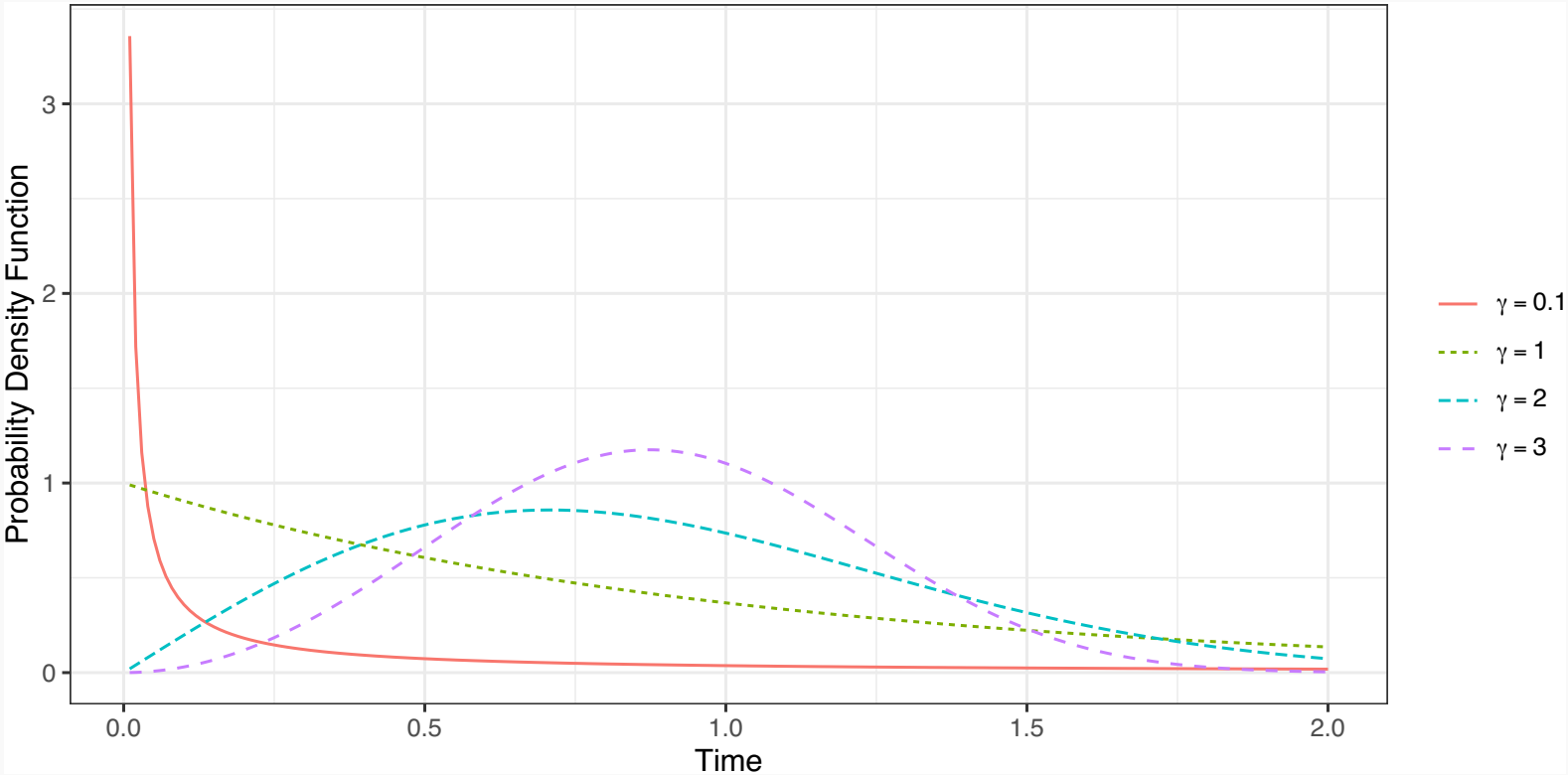


While the exponential model ($\gamma = 1$) has constant hazard, other values for γ provide more flexibility

Weibull Survival Function



Weibull Probability Density Function



Maximum likelihood estimation for the Weibull distribution

The Weibull likelihood for a right-censored sample is

$$
\begin{aligned}
\mathcal{L}(\lambda, \gamma) &= \prod_U f(t_i) \prod_C S(t_i) \\
&= \prod_U \lambda \gamma t_i^{\gamma-1} \exp(-\lambda t_i^{\gamma}) \prod_C \exp(-\lambda t_i^{\gamma}) \\
&= \lambda^d \gamma^d \left(\prod_U t_i^{\gamma-1} \right) \exp\left(-\lambda \sum_{i=1}^{n} t_i^{\gamma} \right)
\end{aligned}
$$

and the log-likelihood is

$$
\log \mathcal{L}(\lambda, \gamma) = d \log \lambda + d \log \gamma + (\gamma - 1) \sum_U \log t_i - \lambda \sum_{i=1}^{n} t_i^{\boxed{\gamma}}
$$

Maximum likelihood estimation for the Weibull distribution

The first derivatives of the log-likelihood are

$$\frac{\partial \log \mathcal{L}(\lambda, \gamma)}{\partial \lambda} = \frac{d}{\lambda} - \sum_{i=1}^{n} t_i^{\gamma}$$

$$\frac{\partial \log \mathcal{L}(\lambda, \gamma)}{\partial \gamma} = \frac{d}{\gamma} + \sum_{U} \log t_i - \lambda \sum_{i=1}^{n} t_i^{\gamma} \log t_i \tag{1}$$

The MLE of λ can be found as the solution to the equation

$$\frac{d}{\hat{\lambda}} - \sum_{i=1}^{n} t_i^{\hat{\gamma}} = 0$$

giving the following in terms of $\hat{\gamma}$

$$\hat{\lambda} = \frac{d}{\sum_{i=1}^{n} t_i^{\hat{\gamma}}} \tag{2}$$

Maximum likelihood estimation for the Weibull distribution

Substituting (2) into (1) gives

$$\frac{d}{\hat{\gamma}} + \sum_{U} \log t_i - \frac{d \sum_{i=1}^{n} t_i^{\hat{\gamma}} \log t_i}{\sum_{i=1}^{n} t_i^{\hat{\gamma}}} = 0 \tag{3}$$

Conceptually, this could be solved by iterative search over γ, for example using `uniroot()` in R, then substituting $\hat{\gamma}$ into the expression for $\hat{\lambda}$ in (2)

Weibull MLE in R

The following R functions evaluate equations (2) and (3) given survival data and γ

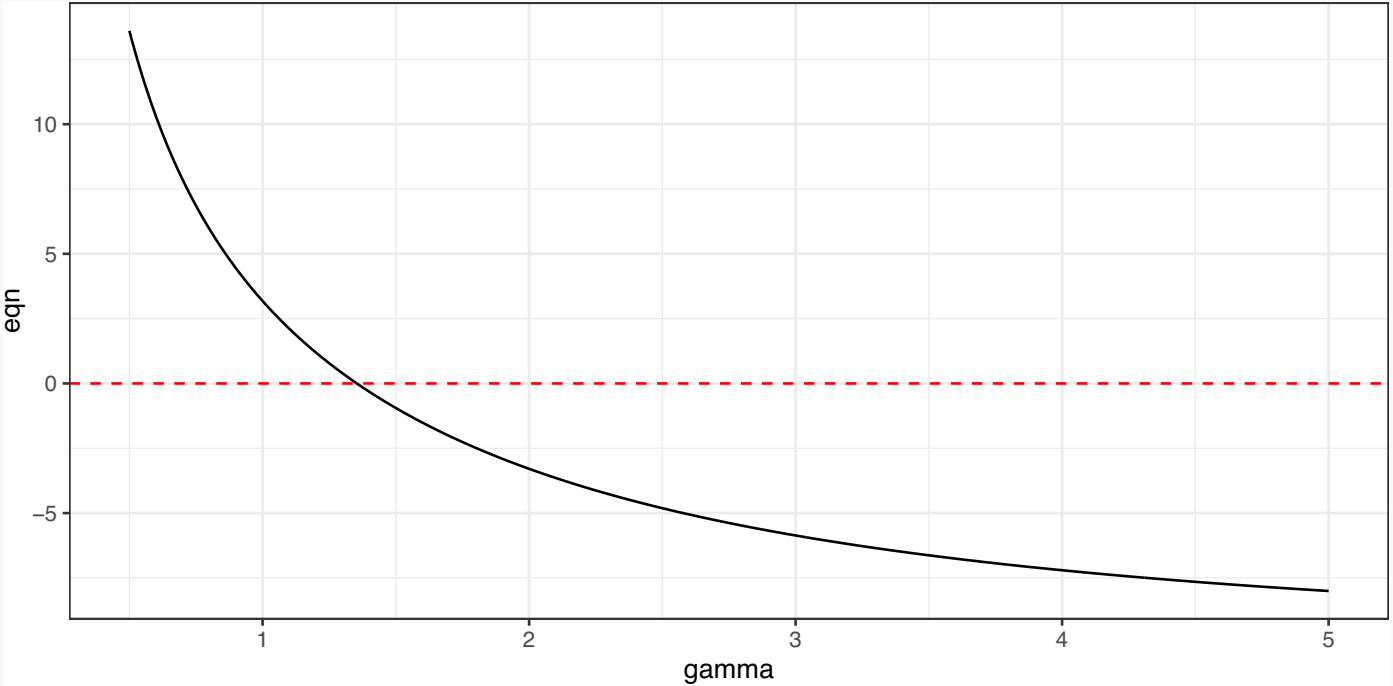
```
lambda.eqn <- function(gamma,events,times){
   sum(events)/sum(times^gamma)
}

gamma.eqn <- function(gamma,events,times){
  d <- sum(events)
  utimes <- times[events==1]
  d/gamma + sum(log(utimes)) -
    d*sum((times^gamma)*log(times))/sum(times^gamma)
}
```

Weibull MLE in R

Using the 6-MP data, we can scan our function `gamma.eqn` across values for γ to find an interval with a root

```
eqn = NULL
for (gamma in seq(0.5,5,0.01)){
  eqn = c(eqn, gamma.eqn(gamma, leukdat1$cens, leukdat1$time))
}
scan <- data.frame(gamma = seq(0.5,5,0.01), eqn)
ggplot(scan, aes(x=gamma, y=eqn)) +
  geom_line() +
  geom_hline(yintercept = 0, color = "red", linetype = 2) +
  theme_bw()
```

Weibull MLE in R



Equation (3) appears to have a 0 in the interval $\gamma \in (1, 2)$

Weibull MLE in R

We now apply `uniroot` for the `gamma.eqn` function on the interval $(1, 2)$ using the 6-MP data

```
gamma.hat <- uniroot(f = gamma.eqn, interval = c(1,2),
                     leukdat1$cens, leukdat1$time)
ls(gamma.hat)
```

```
## [1] "estim.prec" "f.root"     "init.it"    "iter"       "root"
```

```
gamma.hat$root
```

```
## [1] 1.353743
```

Weibull MLE in R

Now we plug in our estimate for γ into λ

```
lambda.hat <- lambda.eqn(gamma.hat$root, leukdat1$cens, leukdat1$time)
lambda.hat
```

```
## [1] 0.008528007
```

The MLE is therefore

$$\begin{pmatrix} \hat{\lambda} \\ \hat{\gamma} \end{pmatrix} = \begin{pmatrix} 0.008528 \\ 1.3537432 \end{pmatrix}$$

MLE using R function survreg

As with the exponential model, we can fit a Weibull model using `survreg`

```
surv2 <- survreg(Surv(time,cens) ~ 1, data = leukdat1, dist = "weibull"); summary(surv2)
```

```
##
## Call:
## survreg(formula = Surv(time, cens) ~ 1, data = leukdat1, dist = "weibull")
##               Value Std. Error     z      p
## (Intercept)  3.519      0.273 12.87 <2e-16
## Log(scale)  -0.303      0.278 -1.09   0.28
##
## Scale= 0.739
##
## Weibull distribution
## Loglik(model)= -41.7   Loglik(intercept only)= -41.7
## Number of Newton-Raphson Iterations: 5
## n= 21
```

$\hat{\mu}$

$\log(\hat{\sigma})$

The R parameterisation is in terms of $(\mu, \sigma)'$, and here $(\hat{\mu}, \hat{\sigma})' = (\underbrace{3.519}_{\hat{\mu}}, \underbrace{0.739}_{\hat{\sigma}})'$

MLE using R function survreg

Estimates for the original parameterisation can be obtained from the relationships $\lambda = \exp(-\mu/\sigma)$ and $\gamma = 1/\sigma$ $\hat{\lambda} = \exp(-\hat{\mu}/\sigma)$, $\hat{\gamma} = 1/\hat{\sigma}$

R gives the covariance matrix of $(\hat{\mu}, \log\hat{\sigma})'$, which can be converted to $\text{var}((\hat{\lambda}, \hat{\gamma})')$ using the delta method

```
c(surv2$coef, surv2$scale)
```

```
## (Intercept)
##   3.5194292   0.7386973
```

```
surv2$var
```

```
##              (Intercept) Log(scale)
## (Intercept)   0.07473057 0.03305811
## Log(scale)    0.03305811 0.07750538
```

Delta method for Weibull

Using the functions $\lambda = \exp(-\mu/\sigma)$ and $\gamma = 1/\sigma$, the delta method covariance matrix is

$$\mathrm{var}\begin{pmatrix} \hat{\lambda} \\ \hat{\gamma} \end{pmatrix} = A\,\mathrm{var}\begin{pmatrix} \hat{\mu} \\ \log\hat{\sigma} \end{pmatrix} A'$$

$$\boxed{\frac{1}{\exp(\log(\sigma))}}$$

where

$$A = \begin{pmatrix} \boxed{\frac{\partial\lambda}{\partial\mu}} & \boxed{\frac{\partial\lambda}{\partial(\log\sigma)}} \\ \frac{\partial\gamma}{\partial\mu} & \frac{\partial\gamma}{\partial(\log\sigma)} \end{pmatrix}$$

$$= \begin{pmatrix} -(1/\sigma)\exp(-\mu/\sigma) & (\mu/\sigma)\exp(-\mu/\sigma) \\ 0 & -1/\sigma \end{pmatrix}$$

$$= \begin{pmatrix} -\lambda\gamma & -\lambda\log\lambda \\ 0 & -\gamma \end{pmatrix}$$

which is evaluated at $(\hat{\mu}, \hat{\sigma})'$ or $(\hat{\lambda}, \hat{\gamma})'$ to get $\widehat{\mathrm{var}}((\hat{\lambda}, \hat{\gamma})')$

Delta method for Weibull

An R function to do the conversion is:

```
weib.conv <- function(params,covmat){
  names(params) <- NULL
  lambda <- exp(-params[1]/params[2])
  gamma <- 1/params[2]
  A <- matrix(c(-gamma*lambda,0,-log(lambda)*lambda,-gamma), ncol=2)
  covmat <- A %*% covmat %*% t(A)
  list(params = c(lambda, gamma),covmat = covmat)
}
weibconv1 <- weib.conv(c(surv2$coef, surv2$scale), surv2$var)
```

The command `%*%` performs matrix multiplication and `t(A)` is the transpose

Delta method for Weibull

```r
weibconv1
```

```
## $params
## [1] 0.008528235 1.353734524
##
## $covmat
##                [,1]         [,2]
## [1,]  0.0001069023 -0.003746487
## [2,] -0.0037464874  0.142036143
```

$\hat{\lambda}$ $\hat{\gamma}$

$$\mathrm{Var}\begin{pmatrix}\hat{\lambda}\\ \hat{\gamma}\end{pmatrix}$$

```r
sqrt(diag(weibconv1$covmat))
```

```
## [1] 0.01033936 0.37687683
```

Confidence intervals for model parameters

The `confint()` function can be used to extract *asymptotic* confidence intervals for the model parameters. For example, here are the 95% confidence intervals for μ and α from the exponential model comparing treatments, and the μ for the intercept-only Weibull model.

```r
confint(survt)
```

```
##                 2.5 %   97.5 %
## (Intercept) 1.7317850 2.587183
## treat6-MP   0.7457452 2.307482
```

```r
confint(surv2)
```

```
##                2.5 %   97.5 %
## (Intercept) 2.983636 4.055222
```

We can also use the likelihood ratio test approach to estimate confidence regions.

Confidence regions for (λ, γ) by LRT

The likelihood ratio statistic for testing

$$H_0 : \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} = \begin{pmatrix} \lambda_0 \\ \gamma_0 \end{pmatrix}$$

$$H_1 : \begin{pmatrix} \lambda \\ \gamma \end{pmatrix} \neq \begin{pmatrix} \lambda_0 \\ \gamma_0 \end{pmatrix}$$

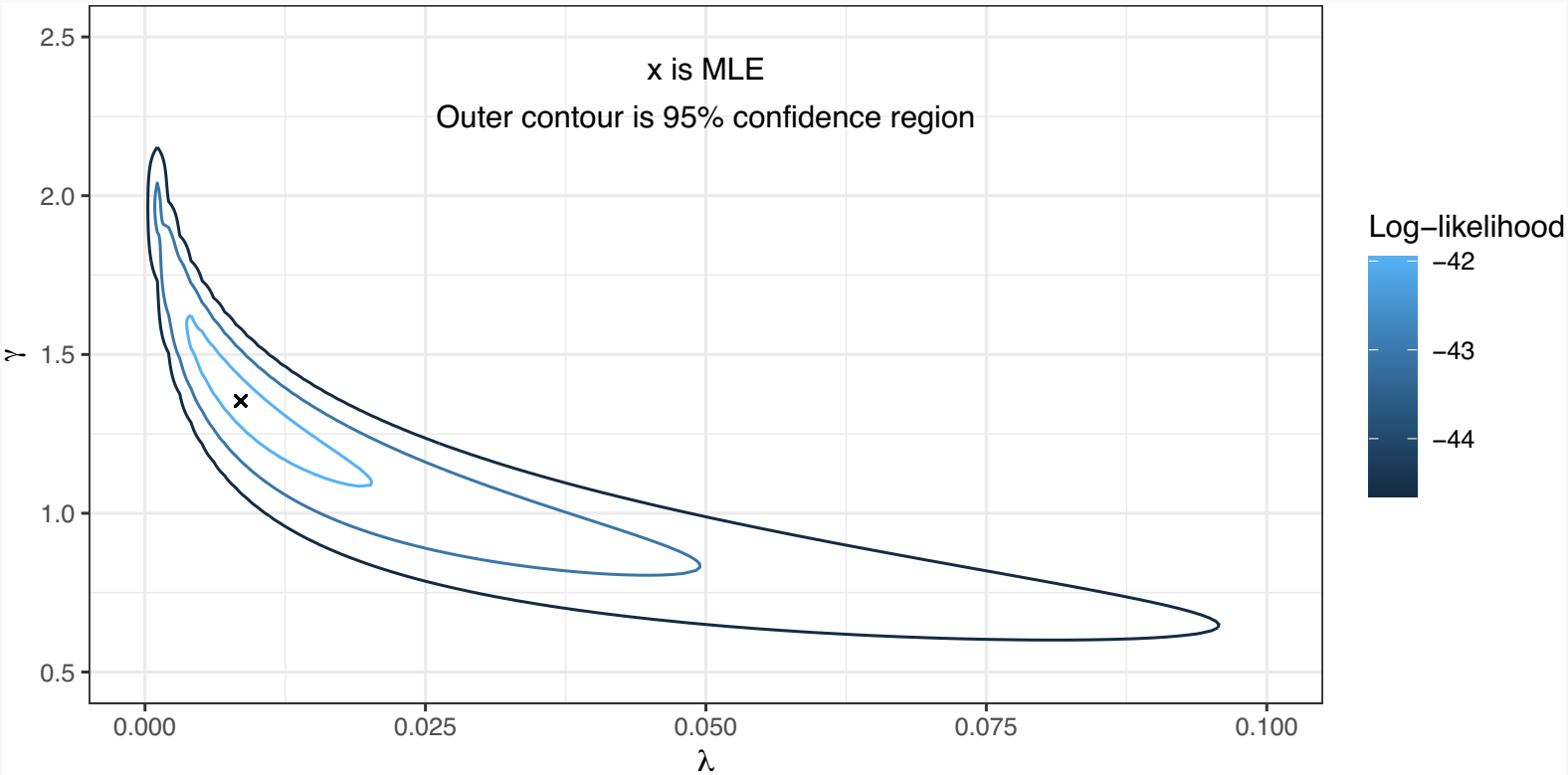
is, under H_0,

$$W(\lambda_0, \gamma_0) = 2(\log \mathcal{L}(\hat{\lambda}, \hat{\gamma}) - \log \mathcal{L}(\lambda_0, \gamma_0)) \approx \chi^2_2$$

So, a 95% confidence region for (λ, γ) consists of all (λ_0, γ_0) such that H_0 is not rejected at the 5% level, i.e.,

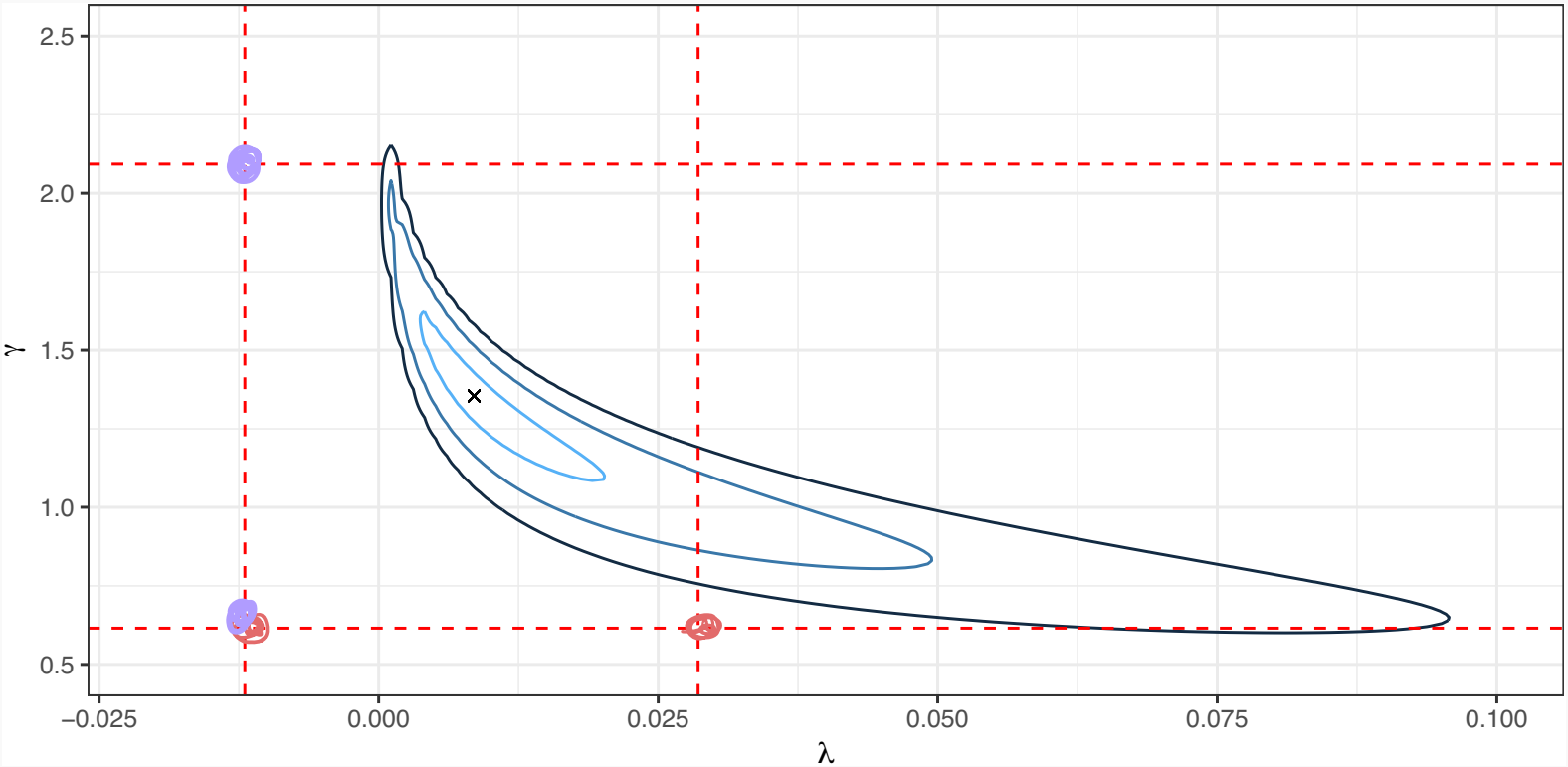
$$\log \mathcal{L}(\hat{\lambda}, \hat{\gamma}) - \log \mathcal{L}(\lambda_0, \gamma_0) \leq \chi^2_{2,0.95}/2 \approx 3.00$$

LRT Confidence Region for 6-MP Leukaemia Trial



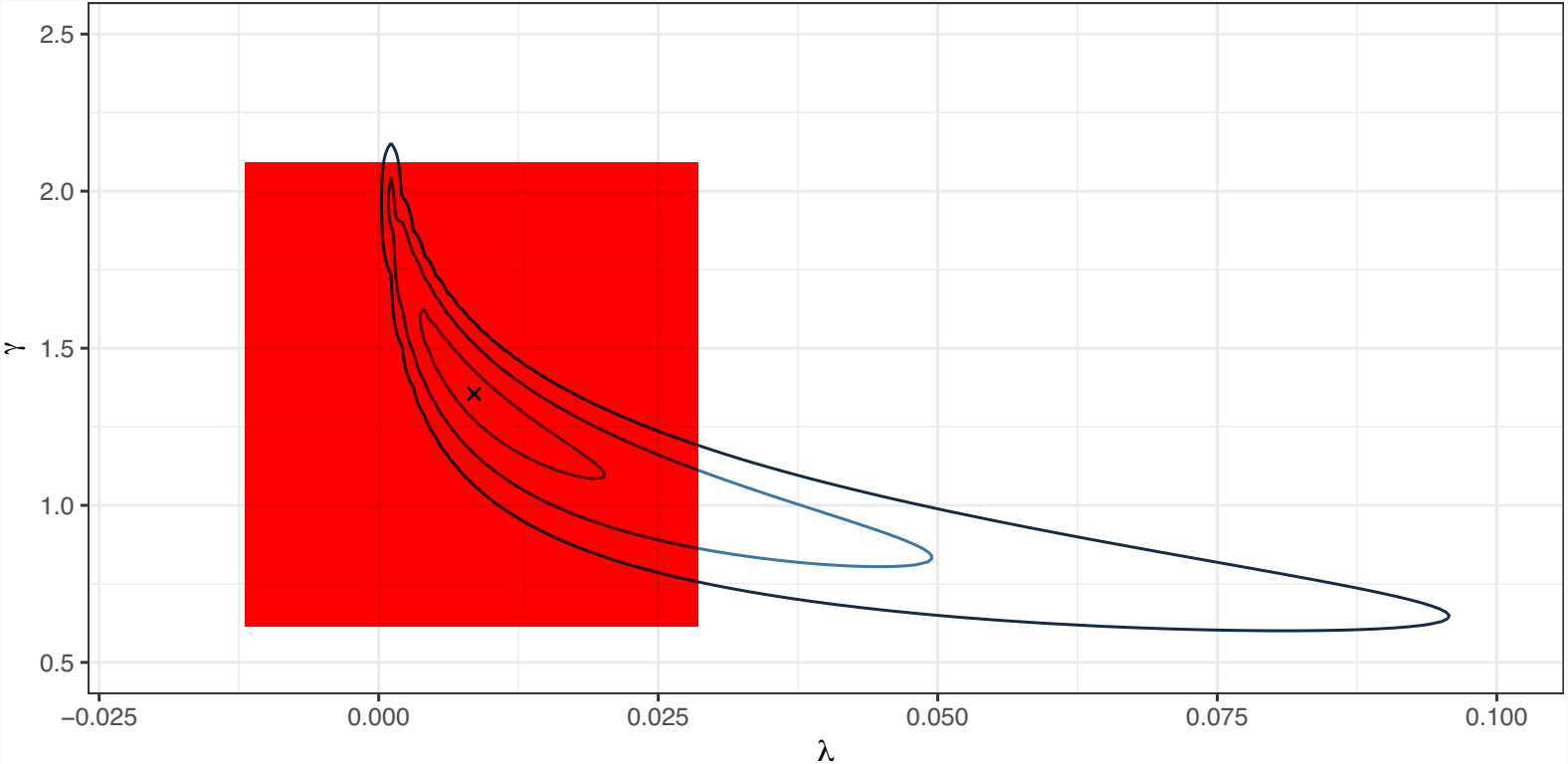
Inner contours represent 25% and 75% confidence regions, respectively

LRT Confidence Region for 6-MP Leukaemia Trial



For a comparison, here is the region created by the bounds of marginal confidence limits

LRT Confidence Region for 6-MP Leukaemia Trial



For a comparison, here is the region created by the bounds of marginal confidence limits

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Confidence Intervals for Central Tendency

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Testing Weibull versus exponential

Testing whether a Weibull (λ, γ) distribution gives a significantly better fit than an exponential (λ) distribution, is equivalent to testing

$$H_0 : \gamma = 1$$
$$H_1 : \gamma \neq 1$$

The likelihood ratio test statistic, under H_0, is

$$W(\tilde{\lambda}, 1) = 2(\underbrace{\log \mathcal{L}(\hat{\lambda}, \hat{\gamma})}_{1} - \underbrace{\log \mathcal{L}(\tilde{\lambda}, 1)}_{2}) \approx \chi_1^2$$

where $\hat{\lambda}, \hat{\gamma}$ are MLE's for a Weibull model with unconstrained γ, and $\tilde{\lambda}$ is the MLE for a Weibull model with $\gamma = 1$ (i.e., MLE for the exponential model)

Testing Weibull versus exponential

For the 6-MP group we get

$$W = \underbrace{2(-41.659 - (-42.175))} = \boxed{1.032} < 3.84 = \chi^2_{1,0.95}$$

so we would not reject the null hypothesis at 5% significance level, and conclude that a general Weibull distribution is not significantly better than an exponential distribution

The `anova` function can be used to compare two nested `survreg` objects

```
surv1 <- survreg(Surv(time,cens) ~ 1, data = leukdat1, dist = "exponential")
anova(surv1, surv2)
```

```
##   Terms Resid. Df    -2*LL Test Df Deviance  Pr(>Chi)
## 1     1        20 84.34976      NA       NA        NA
## 2     1        19 83.31736    =  1 1.032404 0.3095948
```

Testing Weibull versus exponential: Information Criterion

Since parametric models are based on likelihoods, we could also use Akaike information criterion (AIC) or Bayesian information criterion(BIC)

Both cases are based on the -2 times the maximised log-likelihood plus a penalty for each additional parameter

$$AIC = -2\log\mathcal{L}(\hat{\lambda},\hat{\gamma}) + 2p$$
$$BIC = -2\log\mathcal{L}(\hat{\lambda},\hat{\gamma}) + p\log n$$

R has `AIC` and `BIC` functions that could be used. In both cases, the *preferred* model has the smaller AIC or BIC values

Testing Weibull versus exponential: Information Criterion

```r
AIC(surv1, surv2)
```

```
##       df      AIC
## surv1  1 86.34976
## surv2  2 87.31736
```

```r
BIC(surv1, surv2)
```

```
##       df      BIC
## surv1  1 87.39428
## surv2  2 89.40640
```

In agreement with the likelihood ratio test, both *AIC* and *BIC* prefer the exponential model over the Weibull

Other parametric distributions

- The gamma distribution has properties similar to the Weibull. In terms of the Klein & Moeschberger parameterization, the hazard function is monotonic increasing for $\beta > 1$, monotonic decreasing for $\beta < 1$, and is the exponential when $\beta = 1$

- The log-normal distribution has a *hump-shaped* hazard rate, being 0 at time 0, rising to a maximum, then decreasing to 0 as $t \to \infty$
- The log-logistic distribution has a similar hazard to the log-normal. If $\alpha \leq 1$, the hazard decreases monotonically as t increases. If $\alpha > 1$, the hazard is *hump-shaped*
- The log hazard for the Gompertz distribution is linear in time $\longrightarrow h(t) = \alpha e^{\beta t}, \beta > 0$

$\log h(t) = \log \alpha + \underline{\underline{\beta t}}$

The generalized gamma distribution is useful because it contains several other distributions as special cases (gamma, Weibull, exponential, lognormal). Comparison of models can be made by likelihood-based approaches.

Other parametric distributions

In R, the inbuilt distributions can be seen from the help on `survreg.distributions`. There are four basic distributions available: extreme value, logistic, gaussian and t

In addition, there are five derived distributions that are obtained via transformation of the survival time:

- Weibull: extreme value with log transformation
- Exponential: extreme value with log transformation and scale fixed at 1
- Rayleigh: extreme value with log transformation and scale fixed at 0.5
- Lognormal (loggaussian): gaussian with log transformation
- Log logistic: logistic with log transformation

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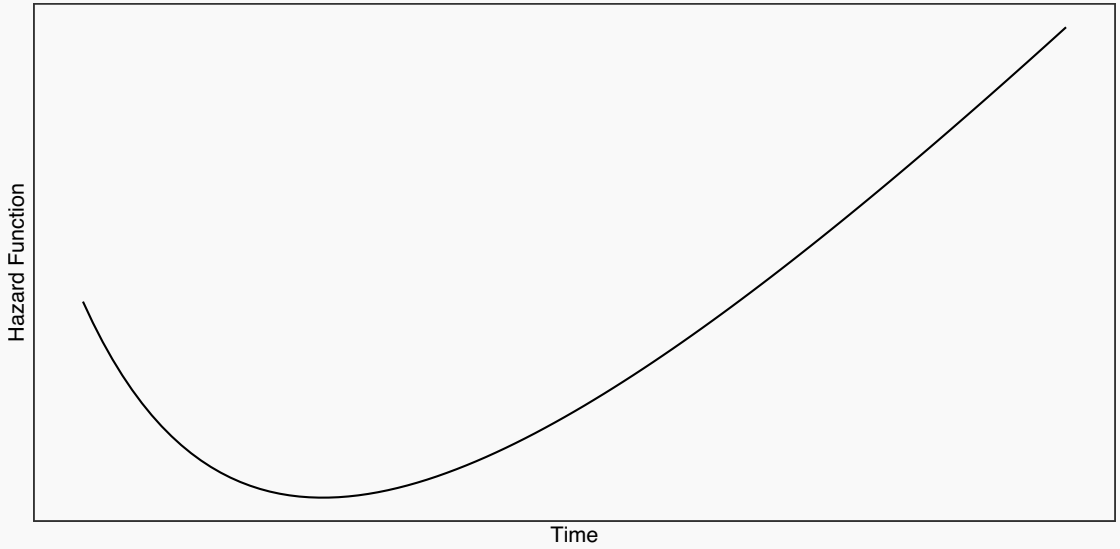
Bathtub Hazard

Bathtub hazard

For human populations followed from birth, a *bathtub* shaped hazard rate is often appropriate

The population hazard follows an initial high risk during the neo-natal period, then decreases to a fairly constant level and finally increases for older ages



Bathtub hazard

Note that here we are talking about the population hazard rate, not the hazard rate for an individual

Odd Aalen makes that distinction below[1]

> *High risk individuals will tend to have a short survival, and the remaining ones will have lower risk, giving a selection over time . . . High risk individuals may be termed "frail"; hence the reason for the popular term "frailty" . . . it is very tempting to interpret the hazard rate as saying something about the development of risk within the individual . . . The simple fact is that the hazard rate is not a pure measure; it expresses a mixture of the individual development and the above-mentioned selection effects.*

[1] Statistical Analysis of Medical Data: New Developments (1998), edited by Everitt and Dunn (Chapter 3: Frailty models, by Aalen)

Bathtub hazard

The bathtub hazard can be obtained by assuming each individual's hazard is comprised of a fixed linear component and an additive gamma random effect or frailty

$$h_c(t|V) = \rho^2 t + V$$

where $V \sim \text{gamma}(\gamma, \lambda)$ which has pdf

$$f_V(v) = \frac{\lambda^\gamma}{\Gamma(\gamma)} v^{\gamma-1} e^{-\lambda v}$$

for $0 \leq v < \infty$; $\gamma, \lambda > 0$

Bathtub hazard

$\exp(-H(t))$

Given this conditional hazard function, the conditional survival function is

$$S_c(t|V) = \exp\left(-\int_0^t (\rho^2 u + V)\, du\right) = \exp(-\rho^2 t^2/2 - Vt)$$

The unconditional (population) survival function is:

$$\begin{aligned} S(t) &= \mathbb{E}_V[S_c(t|V)] \\ &= \mathbb{E}_V[\exp(-\rho^2 t^2/2 - Vt)] \\ &= \exp(-\rho^2 t^2/2)\mathbb{E}[e^{-Vt}] \\ &= \exp(-\rho^2 t^2/2) M_V(-t) \end{aligned}$$

where

$$M_V(t) = \left(\frac{\lambda}{\lambda - t}\right)^{\gamma}$$

is the moment generating function for a gamma(γ, λ) distribution

Bathtub hazard

For population survival function

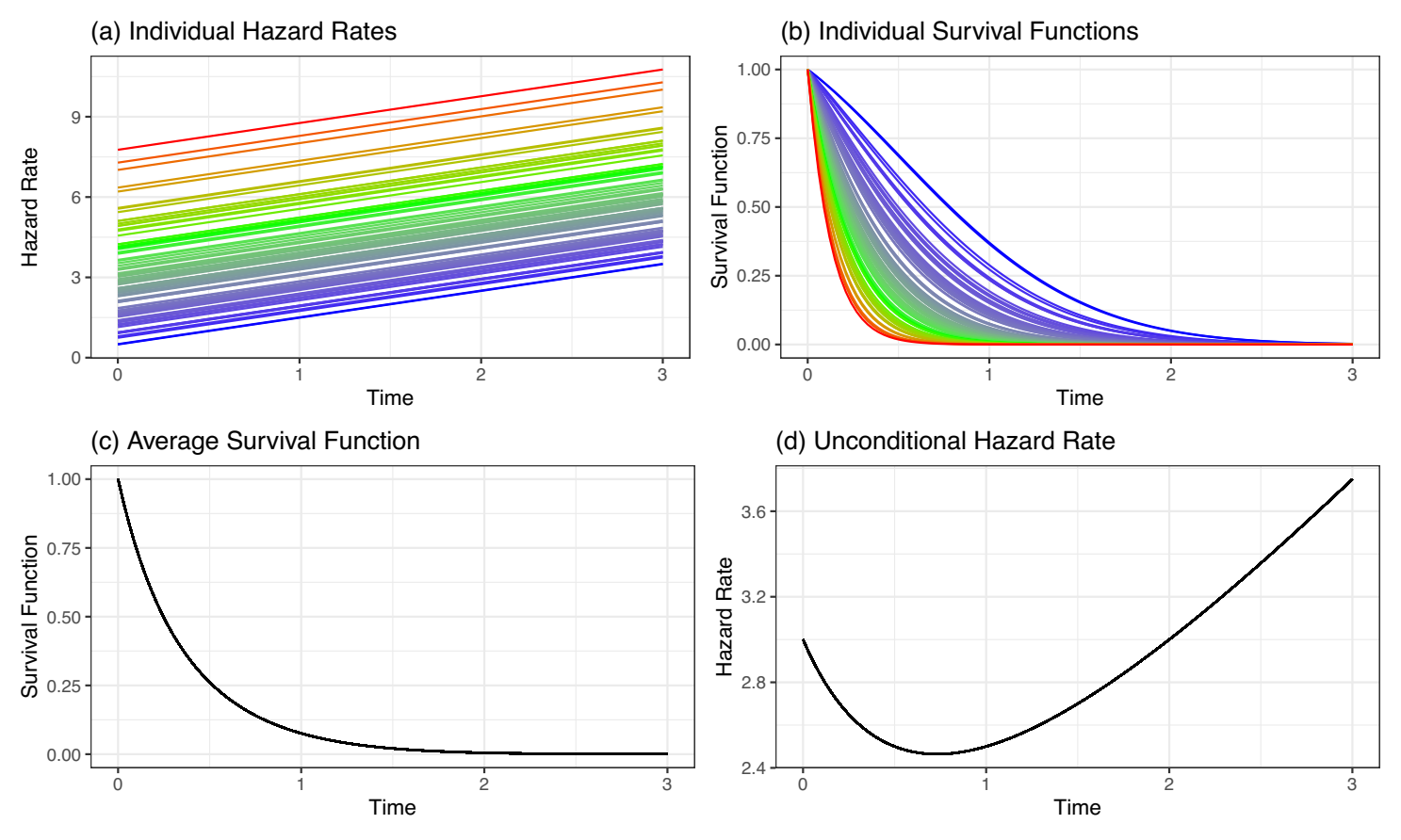
$$S(t) = \exp(-\rho^2 t^2/2) \left(\frac{\lambda}{\lambda + t}\right)^{\gamma}$$

the population hazard function is

$$\begin{aligned} h(t) &= \frac{d}{dt}\left(-\log S(t)\right) \\ &= \frac{d}{dt}\left(\rho^2 t^2/2 - \gamma \log\left(\frac{\lambda}{\lambda + t}\right)\right) \\ &= \rho^2 t + \frac{\gamma}{\lambda + t} \end{aligned}$$

Bathtub hazard

The figure below illustrates how the population hazard function arises.



Bathtub hazard

Here is the R code to produce these graphs for those interested

```
r=1; g=3; l=1; nsim=100; n=100;
times <- 0.03*0:100
vsamp <- rgamma(n, shape = g, rate = l)

dta <- expand_grid(times, vsamp) %>%
  mutate(hc = r^2*times + vsamp,
         sc = exp(-r^2*times^2/2 - vsamp*times),
         sp = exp(-r^2*times^2/2)*(l/(l+times))^g,
         hp = r^2*times + g/(l + times))
```

Bathtub hazard

```r
p1 <- ggplot(dta, aes(x = times, y = hc, group = vsamp)) +
  geom_line(aes(color = vsamp)) +
  scale_color_gradientn(colors = rainbow(3, rev = TRUE)) +
  labs(y = "Hazard Rate", x = "Time",
       title = "(a) Individual Hazard Rates") +
  theme_bw() +
  theme(legend.position = "none")

p2 <- ggplot(dta, aes(x = times, y = sc, group = vsamp)) +
  geom_line(aes(color = vsamp)) +
  scale_color_gradientn(colors = rainbow(3, rev = TRUE)) +
  labs(y = "Survival Function", x = "Time",
       title = "(b) Individual Survival Functions") +
  theme_bw() +
  theme(legend.position = "none")
```

Bathtub hazard

```r
p3 <- ggplot(dta, aes(x = times, y = sp, group = vsamp)) +
  geom_line() +
  labs(y = "Survival Function", x = "Time",
       title = "(c) Average Survival Function") +
  theme_bw() +
  theme(legend.position = "none")

p4 <- ggplot(dta, aes(x = times, y = hp, group = vsamp)) +
  geom_line() +
  labs(y = "Hazard Rate", x = "Time",
       title = "(d) Unconditional Hazard Rate") +
  theme_bw() +
  theme(legend.position = "none")
```

Bathtub hazard

```r
library(patchwork)
p5 <- (p1 + p2) / (p3 + p4)
p5

ggsave('bthaz.pdf', p5, width = 10, height = 6)
```